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Equitable Implementation of the Weatherization Assistance Program in Latine Communities: A Comparative Study of Washington and Tennessee

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1. Introduction

The United States (U.S.) is facing a crucial inflection point in its energy policy, aiming to reduce energy costs and increase energy resilience. Over the past twelve months, electricity prices have increased at more than twice the rate of inflation (1), raising energy bills and placing additional financial strain on low-income families. Improving household energy efficiency for low-income families can significantly reduce their energy burden (i.e., the portion of annual income spent on energy bills), making it an essential part of an equitable energy transition.

The Weatherization Assistance Program (WAP) is a longstanding federal program administered by the Department of Energy (DOE). It was established in 1976 under the Energy Conservation and Production Act, intended to increase energy affordability and efficiency for low-income families (2). Over 45 years later, the WAP has serviced more than 7.2 million homes and continues to serve low-income homes in the U.S. at an annual rate of roughly 35,000 homes (3). The program offers no-cost energy efficiency upgrades (e.g., air sealing and water heater replacements) to homeowners and renters who meet income eligibility requirements (e.g., the household is at or below 200% of the federal poverty line or 60% of the state median income).

The WAP is primarily funded through formula-based grants. Once Congress authorizes and appropriates funds for the WAP, they are allocated to the DOE and dispersed to grantees via block grants. Grantees include the 50 U.S. states, Washington, D.C., federally recognized Tribes, and U.S. territories. In addition to formula-based federal grants from the DOE, grantees may also receive funds from other federal sources, such as the Low Income Home Energy Assistance Program (LIHEAP), or short-term grants (i.e., funds from the 2021 American Rescue Plan Act [ARPA], the 2021 Infrastructure Investment and Jobs Act [IIJA], and the 2022 Weatherization Readiness Fund [WRF]). Lastly, grantees may receive additional funding from non-federal sources, such as utilities, state or local government funds, and philanthropic contributions (4). In FY 2022, nearly $262 billion was allocated to 57 grantees, ranging from $500,000 to $215 million (5). Once the DOE has allocated funds to each grantee, it is the grantee's responsibility to disperse funds to subgrantees and ensure implementation. The main implementers (i.e., subgrantees) of the

WAP are Community Action Agencies (CAAs), primarily public non-profits created under the 1964 Economic Opportunity Act (Economic Opportunity Act, 1964). CAAs implement their services based on federal and state WAP standards, which means that aspects of services, such as which households are prioritized for services, the local workforce completing projects, and the auditing tools (subject to federal approval), can vary between states (3). Every weatherized home is audited by a subgrantee certified Quality Control Inspector (QCI), with additional monitoring for a small subset of homes at the grantee and federal level (3).

Despite being signed into law by Republican President Gerald Ford in 1976 and maintaining bipartisan Congressional support for decades, political scrutiny of the WAP has increased over the last twenty years[1]. After the American Recovery and Reinvestment Act (ARRA) of 2009 increased the WAP's annual appropriation from $230 million to $5 billion over three years, to rapidly expand weatherization efforts in low-income communities (11), more regulations, oversight, and media coverage followed (3,12). After ARRA, a new wave of legislative changes to the WAP emerged between 2020 and 2023, significantly changing the program's appropriation sources, timelines, project scope, and eligibility guidelines as a response to longstanding challenges with the implementation of the WAP (Table 1) (3).

Table 1. Summary of WAP Legislation between 2020 and 2023 (3).

Year	**Program Legislation**	**Summary**

2020	Consolidated Appropriations Act, 2021, Public Law 116-260(13)	● Reauthorizes the WAP through FY2025
		● Increases appropriations for FY2021 to 2025:
		  ○ $330,000,000 for FY2021
		  ○ $350,000,000 for FYs 2022 to 2025
		● Modernizes the definition of weatherization materials by amending Section 412(9)(J) of the Energy Policy and Conservation Act (EPCA) to include renewable energy technologies and other advanced technologies

[1] Support from the Executive branch also fluctuated in recent years. The 2016-2020 Republican administration's budget requests for FY 2018-2020 sought to eliminate all funding for the WAP (6,7,8), whereas the 2021-2024 Democratic administration increased the program's funding, allocating $3.5 billion through 2026 (Infrastructure Investment and Jobs Act, 2021). WAP funding for FY 2022 increased 228% over FY 2021 (5). As the U.S returns to a Republican administration, the FY2026 house passed appropriations for weatherization has decreased from $366 million in FY2025 to $195 million (10)

		● Authorizes DOE secretary to consider non-energy benefits into standards and procedures for the WAP ● Increases administrative funds from 10 to 15% of grant ● Amends section 415(c) of EPCA to allow homes that have been weatherized more than 15 years ago to be eligible for re-weatherization. ● Directs DOE to create a competitive process for financial assistance to support WAP enhancement and innovation
2021	Infrastructure Investment and Jobs Act, Public Law 117-58(9)	● Appropriates $3.5 billion in FY2022 to the WAP, to remain until expended. ● Sets prevailing wage requirement for multi-family buildings (more than 5 units)
2022	Consolidated Appropriations Act, 2022, Public Law 117-103(14)	● Establishes the Weatherization Readiness Fund (WRF) and appropriates $15 million to the WRF for FY2022.
2023	Consolidated Appropriations Act, 2023, Public Law 117-238(15)	● Appropriates $30 million to the WRF for FY2023.

Although prior studies have assessed WAP benefits (16,17), few have examined challenges from the implementor's perspective, except Raissi and Reames (18), who interviewed staff from four Michigan CAAs. Raissi and Reames' (18) work is particularly valuable because they share policy challenges directly from those implementing the program in their community. Their results suggest that in addition to providing myriad social resources to low-income families in their service territories, CAA employees are an under-engaged research partner. Their results identify two main challenges: funding and regulatory constraints to deploying the WAP in their service area (18). Additional scholarship in WAP implementation also suggests that market-related factors related to workforce retention, material costs, and housing stock quality (19,20), as well as social-related challenges, such as public priorities and distrust (21), play a significant role in the program's implementation at the local level.

Given the WAP's longstanding programmatic goals of increasing energy efficiency among low-income families, the program's successful implementation could promote energy justice in the U.S. We define energy justice as the equitable access to and distribution of energy systems and services, along with due process and inclusive representation in decision-making processes, with particular attention to

marginalized or vulnerable populations (22). Prior research suggests energy injustices are more likely in low-income communities and are likely to be magnified by additional forms of marginalization, such as the geography and ethnicity of local communities (23,24). For example, studies indicate that communities with a higher proportion of Black and Hispanic (herein "Latine"[2]) individuals tend to have less access to federal and state energy benefits than other communities (26,27,28). Although many studies discuss the energy injustice faced by Black and Latine households in the US, they tend to refer to racial and ethnic subgroups as homogenous ones (29,30), which is far from the unique lived experiences of Black and Latine individuals. The heterogeneity among Latine populations includes language isolation, geography, socioeconomic status, cultural identity, and country of origin (31,32). These aspects may make Latine families particularly vulnerable to energy injustice by creating barriers to communication with service providers and government agencies and exacerbating the inequitable access to energy infrastructure and energy-efficient housing. For example, a recent study conducted among households located in the U.S.-Mexico borderlands found that 1) homes with Latine-only members, and 2) homes comprised solely of foreign-born non-citizen members had an increased odds of inadequate plumbing, as did households classified as non-English-proficient (33). Importantly, patterns of unequal access are unlikely to be confined to Latine families in remote communities. In the U.S., 12.9% of households are Spanish-speaking households (34), and 19.0% of the population is classified as Latine (34). Thus, Latine households are likely to represent an important demographic for energy efficiency improvements, and those barriers could vary by additional community demographics, such as geography.

Although funding, regulatory, market, and social challenges associated with the implementation of the WAP have been documented separately (18,21), no studies have compared the heterogeneity of WAP implementation across multiple states, which vary significantly by community identity. Further, previous

[2] Although "Hispanic" and "Latine" are often used interchangeably, it is important to note that "Hispanic" is a term coined by the U.S. Census Bureau, intended to classify individuals from Spain or other Spanish-speaking countries (25). Importantly, the term "Hispanic" is criticized for centering Spain, which colonized much of Latin America, and for implicitly excluding people from Brazil and other Portuguese-speaking regions. We opt for "Latine" as a gender- and culturally inclusive term.

studies predate the significant policy changes from 2020 to 2023. Thus, broader research across multiple states that considers all four implementation barriers is needed. To understand the implementation challenges of WAP amid a changing political landscape, our exploratory study has three research aims: 1) establish a relationship with CAA coordinators and gather information about WAP implementation barriers at the community-level, 2) understand how implementation barriers identified in the literature (funding, regulatory, and market) have been impacted by recent policy changes, and 3) explore how community differences (social challenges), such as geography and ethnicity, can further magnify the barriers to equitable program implementation. To do so, we conducted semi-structured interviews with CAA coordinators in Washington and Tennessee. We selected CAAs that serve high proportions of Latine households across different geographies (urban and rural) in the two states of interest. To supplement our interviews, we also assessed sociodemographic patterns using publicly available data and analyzed state-level policies for WAP implementation.

1.1. Study Context

Washington and Tennessee represent a cross-cut of highly variable funding landscapes, political orientations, geographies, and Latine populations in the U.S. According to the U.S. Census Bureau, 44% more Latine adults in Tennessee are non-U.S. citizens with potential resident alien status, compared to Washington (43.2% in Tennessee versus 30.0% in Washington)[3] (34). Latine families are also geographically concentrated differently in the two states: in Washington, the proportion of the Latine population is highest in rural communities, whereas in Tennessee, it tends to be highest in more urban communities (34) (Figure 1). The spatial distribution may stem from different geographic distributions of employment opportunities. Large numbers of Latine adults in both states work in natural resources, maintenance, and construction, but in Tennessee, these jobs are concentrated more in urban areas, whereas

[3] It is important to note that the U.S. Census has repeatedly undercounted the number of Black, Latine, and Native American Individuals in the U.S., while overcounting white non-Latine individuals (35). Tennessee was also recently ranked as the second in the nation with a net undercount rate of 4.78%. (36).

in Washington they are more likely to be found in rural areas (37,38). Washington's population is 7.3 million, and 34% of individuals meet the 200% federal poverty level income requirement for the WAP (34). Tennessee has a population of 6.7 million, and 46% of individuals meet the 200% federal poverty level income requirement[4] (34). Unfortunately, the Census does not report the racial and ethnic composition of the population in this income range.

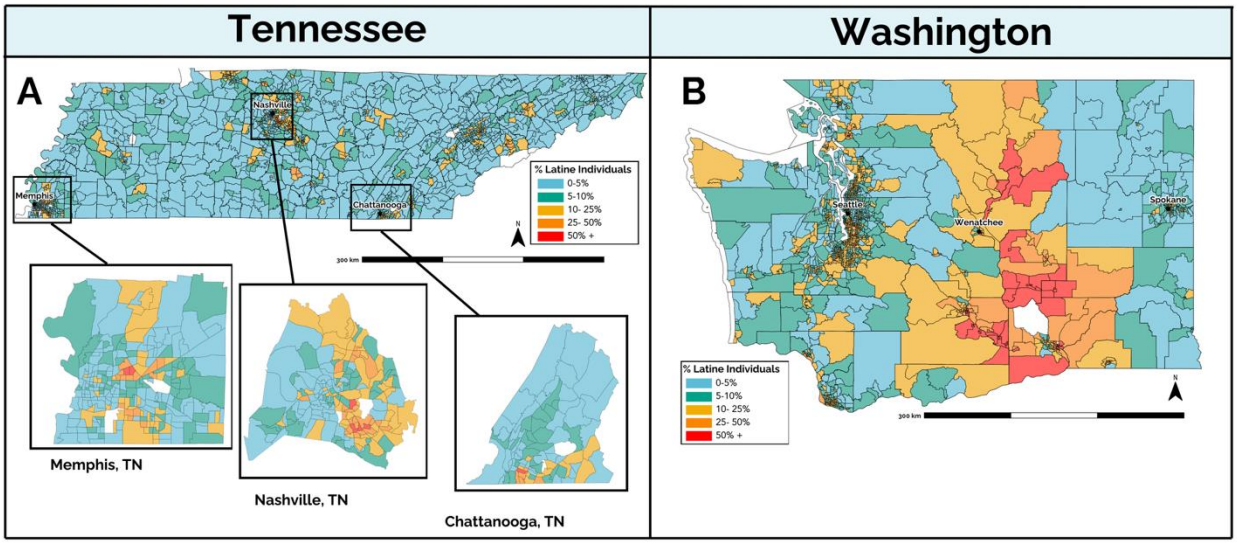


Figure 1. Map of the percentage of Latine individuals in A) Tennessee and B) Washington at the Census Tract level (34).

Although the total populations and percentages of income-eligible people in Washington and Tennessee are relatively similar, the two states have highly different political landscapes and funding structures. Washington received $56.6 million in WAP funds for PY2022, putting it in the third quartile of recipients, whereas Tennessee received $19.4 million, which puts it in the first quartile (Figure 2) (5). For

[4] While the percentage of eligible households in both states is similar, energy burden is experienced differently in both states, due to energy costs and housing stock. Tennessee's average energy burden is nearly twice Washington's (3.0% versus 1.7%) (39). Housing stock in Tennessee and Washington are relatively similar, with both states being primarily comprised of residential homes (i.e., 1-4 units in a structure) (77.8% of homes in Tennessee and 72.4% in Washington) (34). Tennessee has more manufactured homes (8.3%) than Washington (5.6%), and Washington has more multi-unit houses (i.e., ≥5 units in a structure) (21.7% of homes in Washington and 13.7% of Tennessee homes) (34).

implementation, Washington has 29 agencies serving 39 counties, whereas Tennessee has 13 agencies serving 95 counties.

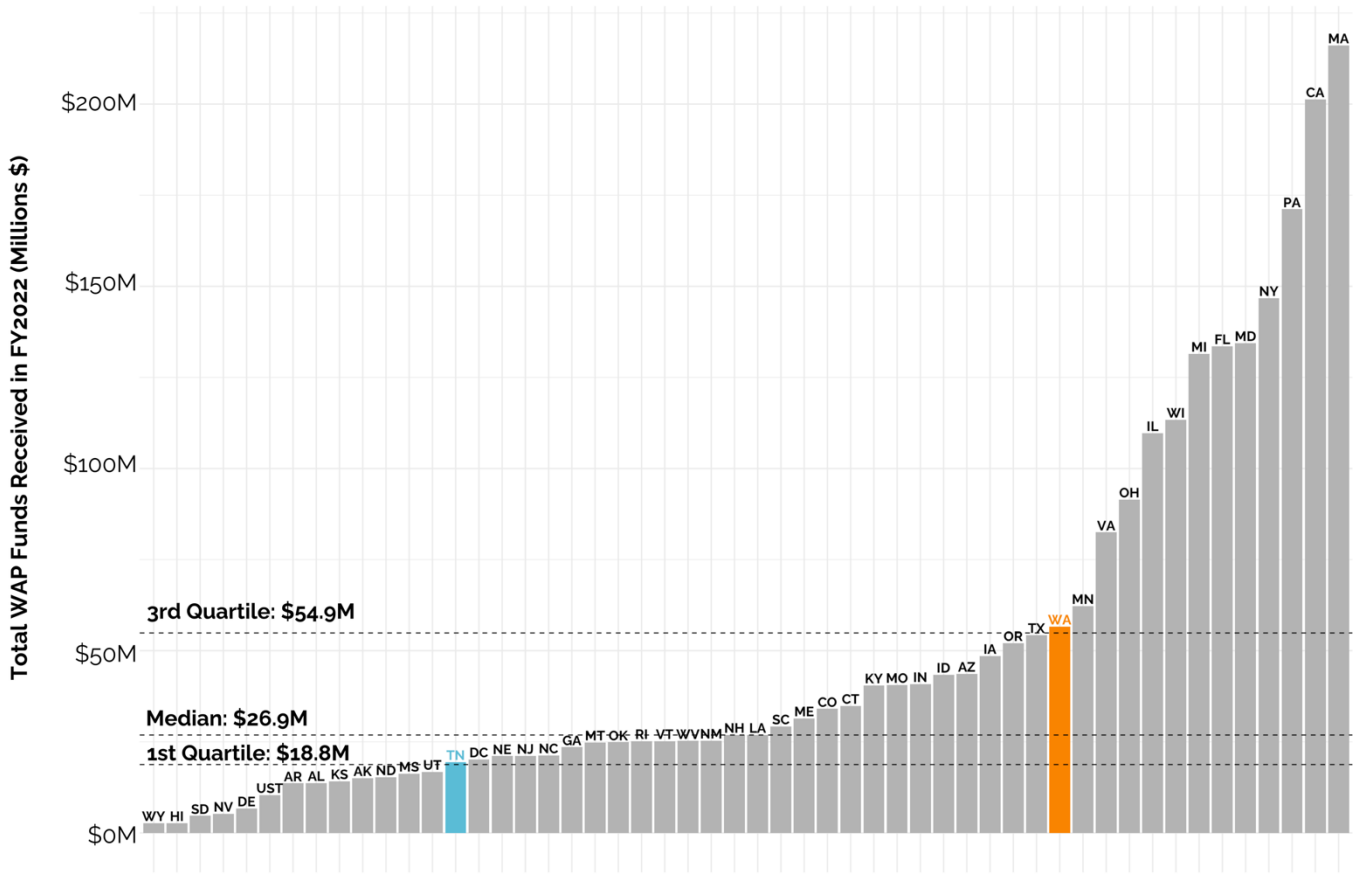


Figure 2. Total WAP funds received in Project Year 2022 to U.S. grantees (5). UST = U.S. Territories.

The most significant difference between Washington's and Tennessee's funding structures is the availability of additional state-level funds. In PY2022, 35.8% of Washington's WAP funds came from non-federal sources (supplemental state-level funds administered by the state legislature) (40), whereas Tennessee did not receive any additional state funds (Figure 3). Both states receive federal funds from the DOE formula and the Low Income Home Energy Assistance Program (LIHEAP). In 2022, Washington and Tennessee grantees both received additional funds from the Infrastructure Investment and Jobs Act (IIJA) (Figure 3). Lastly, Washington was one of 18 grantees who applied and received funds from the American

Rescue Plan Act (ARPA) (5), which provided stimulus funds for recovery from the COVID-19 pandemic (Figure 3), while Tennessee did not receive any ARPA funds.

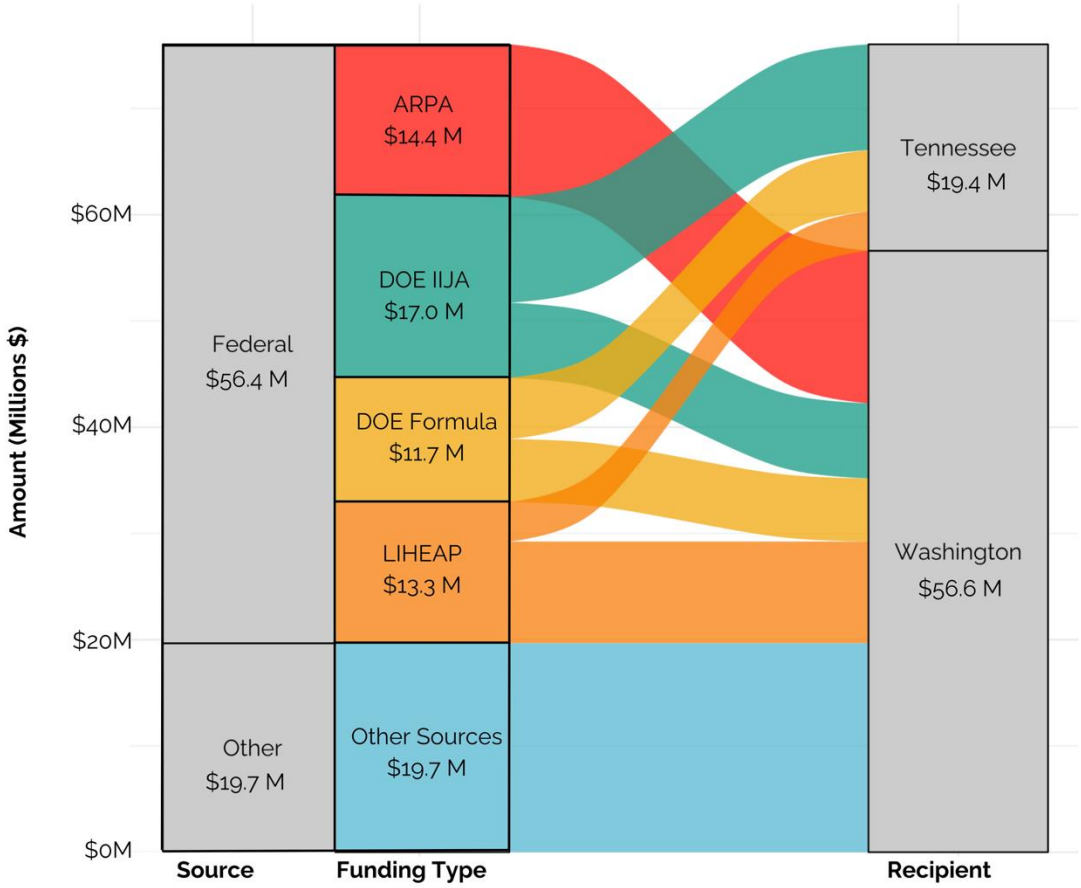


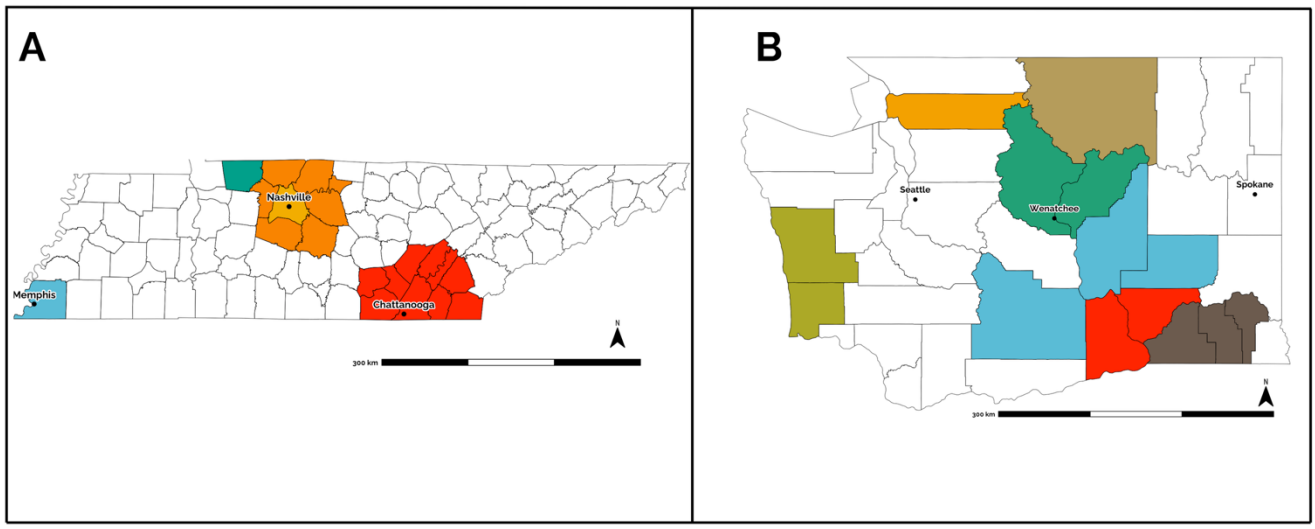
Figure 3. Funding flows for the fiscal year 2022 that Tennessee and Washington state grantees received for distribution to sub-grantee agencies. ARPA = American Rescue Plan Act; DOE = Department of Energy; IIJA = Infrastructure Investment and Jobs Act; LIHEAP = Low Income Heating Energy Assistance Program

2. Methodology and Data

2.1. Participants and Study Procedure

We selected Washington and Tennessee as a purposive sample because we had familiarity and connections to both locations, which have differing Latine populations, geography, political landscapes, and funding structures. Because each state has a limited number of CAAs that provide weatherization services (29 in Washington and 13 in Tennessee), it was important to identify CAAs that represented CAAs

serving Latine populations in rural and urban geographies while ensuring a sufficient sample size. We identified CAAs of interest using 5-year estimates for the percentage of Latine families from the American Community Survey and the percentage of rural communities served by each CAA using the Department of Agriculture's Rural-Urban Commuting Area (RUCA) Codes (37,38). We defined rural as primary RUCA codes 4-10, codes 1-3 were categorized as urban (41,42). In Tennessee, we identified agencies that serve the top 50th percentile of Latine households and the bottom 50th percentile of rural households as interview sites (Figure 4)[5]. Five CAAs or community organizations serve a median of 7.0% of Latine households (Interquartile Range (IQR) IQR1 = 6.8%, IQR3 = 10.6%) and a median of 1.24% of rural households (IQR1 = 0.91%, IQR3 = 9.2%). Additionally, in Washington, we identified CAAs serving the top 50th percentile of Latine households and the top 50th percentile of rural households as interview sites (Figure 4). Eight CAAs serve a median of 25% of Latine households (IQR1 = 21.1%, IQR3 = 34.9%) and a median of 29.4% of rural households (IQR1 = 20.6%, IQR3 = 61.6%).



[5] Because Washington has more than twice the number of CAAs providing weatherization services than Tennessee does, filtering for both Latine and geography indicators in Washington and Tennessee allowed us to narrow the number of CAAs in both states to a comparable number, while ensuring they represented Latine families in different geographies. For instance, filtering solely on CAAs serving the 50th percentile of Latine households resulted in 19 CAAS in Washington, as compared to 6 CAAs in Tennessee. The additional geography filter narrowed the number of sites to a more manageable and comparable subset of CAAs serving similar populations in the state. Moreover, CAAs in Washington's urban areas that serve a high proportion of Latine families tend to serve the same population. For example, there are 6 CAAs that serve the population of King County alone.

Figure 4. A) Tennessee site selection includes five Community Action Agencies serving urban and Latine populations. B) Washington State site selection includes eight CAAs serving rural and Latine populations. Note that agencies are not labeled for anonymity purposes.

Upon identifying 13 CAAs of interest (5 in Tennessee and 8 in Washington), we contacted the lead WAP coordinator at each CAA to request an in-person interview via phone or email using the information listed on the organization's public website. We chose lead WAP coordinators because they were likely to be most familiar with the challenges and strengths of the program, as well as the recent programmatic changes. We conducted interviews between August 2, 2024, and October 24, 2024. Our study was approved by the Vanderbilt University Institutional Review Board (#241100).

2.2. Analysis

All interviews were audio recorded and transcribed using Zoom Cloud Recording, and the transcripts were anonymized following an established protocol, which includes removing all personally identifying information such as names, addresses, city names, proper nouns, and descriptive demographic information (43). We adhered to a semi-structured interview protocol that included background information and questions and probes that aligned with the four implementation barriers identified in the literature (Funding, Regulatory, Market, and Social) (Table 2) .

Table 2. *Abridged version of semi-structured interview guide. Please see Supplementary Table 2 for the unabridged version.*

To start, I'd like to learn more about [CAA] and your role here.	
Theme	**Prompt**
Background	What is your role at [CAA]?
	Can you tell me more about [CAA]?
	What household energy services does [CAA] offer its clients?
Let's move on to discuss your thoughts and experiences about your [CAA]'s funding.	
Theme	**Prompt**
Funding/ Policy	Can you tell me more about how your agency is funded?
	Are there any challenges associated with receiving funding from these sources?
	Do you ever struggle to use all of your funds by the end of the fiscal year?
Now I'll ask about your [CAA]'s eligibility process for receiving household energy assistance.	
Theme	**Prompt**
Regulatory	Can you tell me more about your process for identifying eligible homes?

	What is your CAA's process for household energy improvement?
	How often are households deferred in your service territory?
Great, let's move on to discuss some of the market-related challenges related to [CAA]	
Theme	**Prompt**
Market	Do you ever face challenges serving renters in single- unit homes?
	Great, and how about your experience serving families in multi-unit buildings, such as apartments?
	Are there any challenges specific to your region that make it challenging to provide your services?
Thanks so much for this, the last topic we have is related to the social challenges of household energy efficiency.	
Theme	**Prompt**
Social	Is it challenging to identify and serve families in your service territory?
	Are there any challenges that you've found specifically when working with Latinx families?

The interviewers were two graduate students with interdisciplinary research backgrounds from Vanderbilt University, one of whom was from Washington and had previously conducted qualitative research on residential decarbonization (41), which helped build trust among rural Washington WAP implementors. For each state, interviewers reached data saturation, meaning that new data collected in later interviews often repeated the data collected in previous interviews (44). Upon transcription, auto-generated transcripts from Zoom Cloud Recording were reviewed for quality and, as necessary, slightly edited to ensure they accurately represented the recorded audio.

Prepared transcripts were imported to MAXQDA (release 24.7.0), a software used for qualitative and mixed-methods research for analysis (45). An initial round of open coding was conducted using both deductive and inductive codes. Codes informed by previous literature, such as challenges (e.g., multiple funding sources and evolving regulatory limitations), were identified deductively (18,41), whereas new and recurring themes (e.g., immigration status) were identified inductively (46). Subsequent rounds of thematic coding were conducted to consolidate and clarify themes and address uncertainty (46). In the final coding round, axial codes were created to categorize the codes into the four implementation themes (funding, market, regulatory, and social) and to indicate whether implementors framed the statement as a challenge, a strength, or a desire (Supplementary Figure 1). We also utilized state-level tags to examine differences between states.

3. Results

Among the 13 CAAs contacted, nine representatives responded and agreed to be interviewed. Our interviewees represent five CAAs in Washington and four CAAs in Tennessee, collectively serving 27 counties. Interviews were predominantly in person at the CAA of interest's office; two were held via Zoom due to unforeseen circumstances. Our study revealed common themes across both states, and a small subset of state-specific themes (Figure 5).

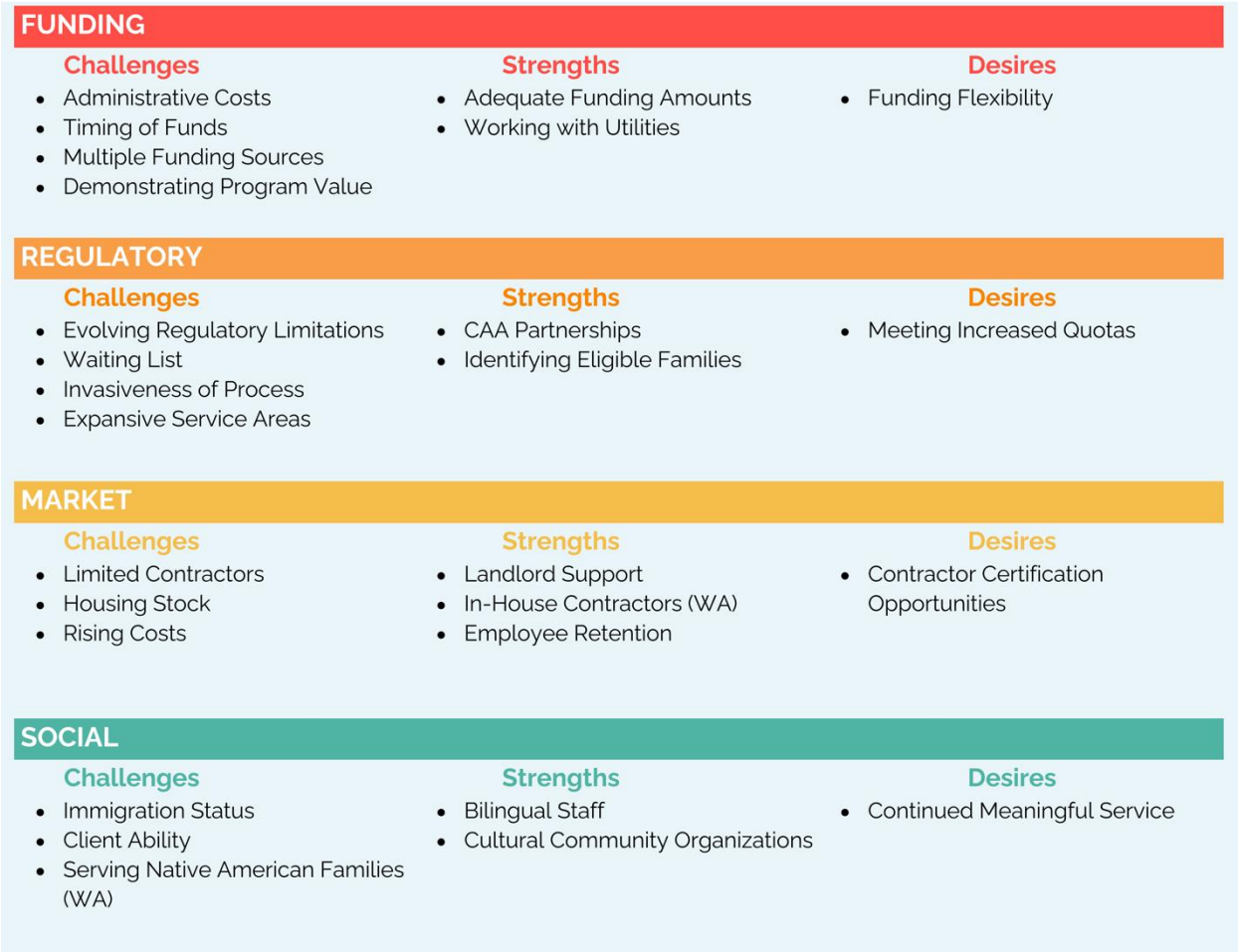


Figure 5. Summary of challenges, strengths, and desires shared by Washington and Tennessee Weatherization Assistance Program implementers, categorized by funding, regulatory, market, and social themes.

Interviewees reported working for agencies that served between one and ten counties and weatherized between 15 and 55 units annually. An important difference was that three Washington agencies had in-house construction crews, whereas Tennessee did not have any.

3.1. Funding

3.1.1. Challenges

One of the biggest challenges in funding was the 15% limit on funding administrative staff (i.e., individuals responsible for paperwork and data entry), which was recently increased from 10% by federal guidelines in 2020 (Table 1). Although administrative cost limitations are common among similar programs such as LIHEAP, coordinators in Washington and Tennessee communities that served populations with inadequate housing conditions or lacked in-house construction teams frequently cited challenges associated with administrative limits, due to the additional time it took CAA employees to identify homes that were eligible for the program and process bids for external construction teams and QCIs. When asked if there were any challenges associated with receiving funds from federal and state sources, an individual in Washington shared,

> "I can't hire enough people to go through the funds that I have the potential to go through. For example, we gave back probably close to a million dollars back to the state because we couldn't--We just didn't have the manpower to--process the applications, to do the math, to do all the things that need to be done... and we have the money to do it… this is a community that doesn't need us to be giving back a million dollars."

Interviewees in both states cited challenges with the timing of funds, due to delays at the state level or in coordinating jobs. Although the challenges associated with federal funding distribution are not uncommon (47), the funding landscape in Tennessee was especially complex because it runs on a reimbursement structure, meaning that the CAA is responsible for paying for energy efficiency improvements upfront (e.g., material costs and contractor and QCI payments), before the grantee will reimburse the agency for the project (48). An individual from a Tennessee agency shared,

> "We have to float the contractor their pay … but some of our jobs go for…over 20 grand. So that's money that you have to float out to the contractors as well as still pay your employees."

Another challenge associated with funding in both states was combining multiple funding sources. Importantly, this was more frequently mentioned by Washington interviewees. For example, a Washington employee shared,

> "Weatherization funding is extremely complex. And we can have 5 or 6 funds in one home, depending on the house and how it's heated, and what, how much money we have, what needs what, and when a contract date is going to end."

Lastly, interviewees in both states mentioned challenges associated with demonstrating the value of weatherization to community members and program evaluators, who might be less familiar with the program and its added benefit to communities. Although Tennessee implementors are only subject to the QCI process, Washington implementors receiving additional funds (e.g., state or utility funds) can be subject to additional evaluators. For example, one Washington implementer described their experience with a utility evaluator,

> "It's been 10 years ago that a gentleman…evaluated me and our program… [he said], 'on average, you're saving 6% across the program, that's all you save us… these two jobs are using more energy than before you left them'… So I went and followed up, and one of them was in a very rural community… and people were so comfortable, they moved more family members in… and they had grandma plugged into an RV just outside… And then another one, [the family] had transformed it after we got done, into a duplex and they're renting the basement out to their family. So you could judge our program on energy savings. But it's not accurate. It's really, it's a bigger picture. And it's a community. It's health, all kinds of other things."

3.1.2. Strengths/Desires

One of the topics that interviewees in both states most frequently discussed was the addition of Weatherization Readiness Funds. Interviewees often expressed gratitude for the increased flexibility to tackle health and safety challenges that would typically be postponed. A Tennessee coordinator shared their experiences with this.

> "We've luckily… got certain funds set aside from THDA [Tennessee Housing Development Agency] that are what we call weatherization readiness funds, where I've been able to bid out jobs…we've been able to get people … turn around and go right back in and complete the energy audit and get them help. So that has been a game changer for… deferrals."

Some Tennessee interviewees felt limited by the eligibility criteria for Weatherization Readiness Funds that the grantee had provided agencies, which require subgrantees to evaluate the home, quote the repair costs, and email photos and the WRF funding request to the grantee for approval (49). Interviewees suggested clearer guidelines and more interaction with state coordinators.

Another strength interviewees in both states identified was working with utilities to maximize services offered and fill funding requirement gaps. Washington interviewees discussed receiving additional funds directly through their local utilities, which was often their most flexible funding source. Tennessee interviewees reported that working with local utilities allowed them to refer families that do not meet eligibility requirements. Some interviewees in Tennessee also mentioned working with the Tennessee Valley Authority's (TVA) Home Uplift program, a TVA-funded energy efficiency program for low-income customers in the service region. While interviewees generally felt the program was complementary, some shared that it made acquiring certified contractors in their areas more challenging. Additionally, one interviewee in Washington shared that gas utility reimbursements for customers in their region were more generous than electric companies, which may be counterproductive for Washington's ambitious climate goals.

The greatest funding desire that interviewees in both states identified was additional flexibility to ensure they could meet the unique needs of their communities and decarbonize homes. Interviewees discussed replacing decades-old electrical systems to accommodate new technologies, switching fuel sources from oil to electricity, and providing families with safe, comfortable, and affordable housing. One coordinator in Tennessee shared their experiences with this,

> "It really has become more of a battle of keeping people in their homes. So going in and being able to…make those repairs…that will actually keep somebody in their home. So they're not homeless… [and] able to actually live comfortably… that's what it's all about."

3.2. Regulatory

3.2.1. Challenges

The most commonly discussed regulatory challenge among Tennessee and Washington interviewees was the evolving combination of federal and state regulatory requirements for the program. Interviewees in both states discussed the challenges of meeting specifications for acquiring the correct inventory (i.e., the "Build America, Buy America" products pursuant to Title IX of HR 3684), managing the competitive bidding process for hiring contractors, and ensuring clients meet the necessary eligibility requirements. Washington

interviewees also discussed the challenges of meeting standards for prevailing wage, which is a state-level requirement for the Washington WAP (50). One coordinator in Washington expressed their frustrations with this,

> "Some of the specifications and standards…all have good intentions, but they've layered on top of themselves over time to make… almost undeliverable products. To where the specifications, and the regulations, and the process has become almost unmanageable."

Interviewees often discussed the challenges related to the prioritization process of their agencies' waiting lists. Although grantees are given flexibility to define their own vulnerable groups for priority (e.g., both states prioritize elderly, disabled, and dependent children in low-income households, while Washington also prioritizes low-income Native American households, and Tennessee does not), the process tends to be the same. CAAs are instructed to assign points additively based on whether the low-income household has a vulnerable member; additional points are added for energy burden and high residential energy consumption (48,50). Thus, the prioritization queue often shifts based on the status of other clients, resulting in feelings of confusion or frustration for clients due to the lack of transparency surrounding the service timeline. One WAP implementer in Tennessee shared,

> "If we could change something about eligibility. I would personally scrap priority points entirely. I think the intention is, is noble, like we all want to serve the most vulnerable, but I just feel like it does not serve the clients… what the client wants to know is, 'where am I on the wait list?'… Somebody could apply today and have 75 points, which is like the maximum, and they're gonna jump ahead a lady that's been waiting since 2017 because she only has 5. So what is the point of the waiting list?"

In addition to the challenges associated with the prioritization process, interviewees in both states also discussed that the process for qualifying eligible families can feel invasive for some clients. They shared that the paperwork requirements were extensive, and the improvement process can feel intrusive, because holistic energy efficiency retrofits require auditors and contractors to work in the home for extended amounts of time. For example, one coordinator in Tennessee shared,

> "It's probably one of our biggest obstacles that our county offices see because people want to come in and give you A, B, C [documents], but we need D through F, and they're like, 'Oh, but I brought everything'. And then it's a phone call to somebody higher up. And you're like, 'no, this is, this is a state thing…we're just administering what the State has given us."

The final challenge coordinators in both states mentioned was serving clients in expansive service area territories. In both states, implementers mentioned sometimes spending more than two hours traveling from their agency to a client's home. In Washington, longer travel times were often associated with living in more rural areas, whereas Tennessee coordinators often faced much larger service territories.

3.2.2. Strengths/Desires

Implementers in both states discussed the value of strong partnerships with other CAAs, local nonprofits, and state-level coordinators to better serve their clients, share ideas on how to best meet regulatory specifications and standards, and resolve regulatory uncertainties. One Tennessee coordinator shared,

> "When I started, I wanted to reach out to other agencies for … best practices. To say, 'look, you all have been running it much longer than I have been here. So how do you handle contractor engagement? How do you handle staffing up? How do you handle capacity issues and deferrals?' And so I just spent a little bit of time like calling them up and just having these meetings and conversations about it. So I feel like it's grown our relationship that way because we share similar struggles."

Several implementers in Tennessee mentioned frequent turnover among state-level coordinators, which sometimes made these partnerships more challenging.

Another strength discussed in both states was identifying eligible households. Interviewees often discussed leveraging partnerships with other non-profits and utilities for client referrals, attending public events such as farmers' markets and community health fairs, advertising in newspapers and radio, and using social media platforms. Interestingly, interviewees often shared that "word of mouth" from clients previously receiving the service was the most valuable approach for identifying new eligible clients. One implementer in Washington shared,

> "[One employee] went to different mobile home parks… and talked about our program, where we've done a little bit of work already, and there were some… happy clients and customers that happened to be there, who could speak to their experience, you know 'They did my trailer, and it's wonderful now', and we ended up with gosh, half a year's worth of work."

Importantly, interviewees in both states mentioned that they faced challenges in convincing eligible elderly clients to take advantage of their program. Often, elderly individuals would express that the funds should be given to someone else who needs them more. For example, in Tennessee, one coordinator shared,

> "It's hard to get clients that want to apply… because most of them in the county, they're older and their mindset is… 'I got it, give it to somebody else that needs it,' you know, and you might need it."

A frequently expressed desire in both states was the ability to meet increased production goals set by state-level coordinators. In most cases, implementers had seen their production goals double in the past year, and they were eager to serve more individuals so long as the guidelines and restrictions did not keep them from doing so. For example, one coordinator in Tennessee shared,

> "It's [increased quotas] a good problem to have. It gives us something to strive for, to help more people."

3.3. Market

3.3.1. Challenges

One of the most frequently discussed challenges interviewees in both states shared was the lack of certified contractors necessary for weatherizing clients' homes. According to most interviewees, the major contributors to this issue are the extensive paperwork associated with the bidding process, lengthy certification requirements, and competing contractor opportunities. In Washington, interviewees who did not have in-house contractors tended to discuss contractor shortages as a result of extensive paperwork and the prevailing wage requirement in their state. One Washington coordinator shared,

> "The bottom line is you have to pay prevailing wage and you have to file intents and affidavits on every job that you do, and you have to follow their [the State's] …rules, and a lot of the contractors that are in [redacted region], there's plenty of work and construction going on here."

In Tennessee, coordinators voiced similar frustrations about a limited workforce, although they encountered somewhat different challenges. One frequently cited obstacle was the difficulty that contractors face in obtaining government certification. Unlike Washington, where local certification is available in-state,

contractors in Tennessee are required to obtain certification in Virginia due to state requirements. This mandate places additional financial burdens on contractors, who must be prepared to travel out of state for certification and forgo wages during the certification period. Additionally, agencies in Tennessee compete with other organizations for certified contractors offering similar programs. Most recently, the TVA's Home Uplift program has increased the strain on an already limited workforce, because Home Uplift tends to have fewer contractor requirements. One Tennessee coordinator shared,

> "When I talked to our contractors which left us… They said, 'Why would I bid on this home and go through all of this when I can bid on this bigger place, and it's less stress?' … And it's just not us. It's a lot of the agency contractors, left and started bidding on those jobs. And like, for the last 2 years, we have not got our houses done because of the contractors ."

Interviewees also frequently mentioned challenges associated with housing stock in their service areas. The most common challenge associated with housing stock among Washington and Tennessee coordinators was multi-family units. In Washington, several interviewees discussed the limited availability of eligible multifamily homes in their rural service territories. These limited numbers were sometimes considered a challenge because servicing multi-family homes often costs less per unit, so that agencies can serve more families with their budgets. One coordinator in Washington shared,

> "We do [serve multi-family homes] if we have them to serve. But all the units that we have here in our county have all been served by us before, or have been done since then, with other types of state funding."

In Tennessee, coordinators frequently mentioned that the state had recently reclaimed sub-grantees' ability to serve multi-family homes greater than five units (48). Some interviewees mentioned that this has increased the strain on contractor availability and has also made meeting weatherization goals more challenging. A Tennessee coordinator shared their thoughts on these changes,

> "Tennessee, recently… told us we could not work on multifamily buildings…And that used to be one of the easiest ways for us to hit our unit goal… and I think that their plan was that that was going to take a burden off of us…But unfortunately, that hasn't been the case…what's happened is that they've taken existing contractors …So it's actually removing more contractors away from us."

The challenges associated with weatherizing manufactured homes (i.e., mobile homes) were also commonly discussed in both states, although Washington coordinators tended to discuss them more often. This is likely because rates of manufactured housing stock were much higher among Washington CAA service territories interviewed than Tennessee CAA territories (median of 11.3% housing stock among Washington CAAs, versus 3.8% among Tennessee CAA territories (34)). Although coordinators often shared the challenges of investing significant amounts of funds into manufactured homes that likely have a low assessed value, coordinators were quick to state that the investment had meaningful value to low-income community members and would remain as a critical affordable housing unit in the community. One coordinator in Washington shared,

> "I have no problem with mobiles. Some people do, because when you look at the assessed value of a mobile home, a lot of the times, the assessed value is…so low in comparison to the amount of money that you're putting into weatherizing. But like I tell other people around the state… I have absolutely no problem spending the amount of money… it's a home just like anybody else's home. And if done properly, you know, it's gonna be there for a long time. And 90% of the time, it's gonna stay low income… and the person living in that home, they can't afford to do anything else. They couldn't afford to pay for rent or to pay for anything different. So there's that value there."

It is important to note that manufactured homes are more vulnerable to extreme heat compared to other types of housing. When combined with poor insulation and economic constraints related to cooling, residents are more likely to face higher energy burdens and health risks (51).

Several coordinators in both states also discussed health and safety challenges associated with housing stock that may not be covered by Weatherization Readiness Funds, which were hard to turn away from. For instance, coordinators shared challenges related to roof repairs, deteriorating floors, asbestos, mold, and vermiculite. One coordinator in Tennessee shared,

> "This guy had … big old huge holes in his ceiling…I'm sitting there, you can see the sky… I'm like, 'and when it rains?'… [he said,] 'I just put an umbrella and stick it through that and open it up'… I'm like, 'you know if it's wet, mold is gonna come in, you know, in this floor'."

Lastly, coordinators in both states frequently mentioned the challenges of rising costs of labor and materials. Moreover, they often shared that their rising costs were felt most after the COVID-19 pandemic and were increasing at a rate that policies and regulations could not keep up with.

3.3.2. Strengths/Desires

Despite the market-related challenges, coordinators in both states frequently mentioned the interest and willingness of landlords to collaborate. However, expectations for landlords in both states are different. In Washington, landlords are expected to contribute in some aspect of the process, either financially or through involvement in the process. In contrast, landlords in Tennessee have no expectations for receiving services in their properties. One coordinator in Washington mentioned,

> "It [weatherization] works better when the landlords, you know, have a buy-in. And there's also a caveat in our funding that, says the landlord has to contribute-- it doesn't say what… It could be that they contribute labor…They could help us do stuff, they could give us a thousand dollars, I mean, we leave it wide open to them. So, landlords… that are proactive and interested in saving energy and making a better place for the residents to live… it's a little bit more work for us, but not a lot."

One strength that was only relevant for Washington coordinators was the presence of in-house contractors. When applicable, in-house contractor crews in Washington removed the burden of external contractor shortages, increased the interviewee's ability to coordinate several jobs at once, and ensured that jobs met restrictive federal requirements. Implementers in both states frequently shared aspirations for in-house contractors. One coordinator in Tennessee shared,

> "We have started the process of trying to develop in an in-house crew to where [CAA] would employ the crew that is completing those weatherization measures… eventually what we want to do is get to a point where all the work is completed in-house so that we can move faster."

Lastly, interviewees in both states shared the value of maintaining CAA employees for years and even decades. Seasoned employees ensured coordinators were able to address regulatory changes and funding challenges.

The most frequently expressed desire among coordinators in both states was for accessible and affordable local workforce training. Expanding training opportunities would incentivize contractors to consider weatherization as a career, increase the availability of qualified contractors, and improve the likelihood of meeting increased production goals. One Tennessee coordinator shared,

> "They [prospective contractors] need exposure to know if they even want to work in this industry. But if you upfront slap them with a list of trainings they're like, 'oh, well, I probably don't want to do this'. You don't even get a taste for weatherization…we want to support them because ideally, we want to increase the skill set of the workforce here for weatherization. We already know there's a great need here, but we don't have many solutions to serve this need. So we need people, local people trained in this, and providing services to other citizens here."

3.4. Social

3.4.1. Challenges

When asked about the challenges of reaching and serving eligible families, interviewees in both states noted the federal eligibility limits linked to immigration status. Importantly, these challenges were mentioned more frequently in Washington than in Tennessee. For instance, interviewees in Washington often pointed to the policy change that took effect in July 2016, revising the eligibility limits to tighten the definition of documented immigrants or "qualified aliens" (i.e, immigrants with documentation to prove they have the status of legal permanent resident, refugee, special immigrant, asylee, parolee, deportation withheld, conditional entrant, battered spouse or child of U.S. Citizen or permanent legal resident, Cuban or Haitian entrants, U.S. military veteran, victim of trafficking, or American Indian born in Canada) (50). After this change, interviewees stated that the immigration status limits became a deterrent for Latine families with members who had precarious immigration statuses. One interviewee in Washington shared,

> "I've lost a…multifamily weatherization job, because commerce requires citizenship documents for all of the people who are receiving weatherization services in spite of whether or not they're living in a multifamily unit or not. which I think is a travesty. And here's why. Nobody is making any money or generating any wealth off of living in an apartment. So by telling a, an agency that we cannot update or weatherize a low income situation…simply on the basis of the fact that they allow people who don't have the right piece of paper to receive this service, is hurting the community. And I would even posit that there's some institutional discrimination going on with that, and maybe not towards the individual--institutional discrimination-- but in the institutional discrimination towards the community."

Interviewees in both states mentioned that only one household member needed to meet the eligibility limits, so households including both undocumented immigrants and documented immigrants (e.g., if parents are undocumented but a child was native-born) could still be served.

Another social challenge frequently referenced by interviewees in both states was client ability. Although Weatherization Readiness Funds often helped to improve structural challenges associated with a client's home, several interviewees referenced instances where clients were unable or unwilling to improve their homes in preparation for receiving the services. Often, these limitations could be ascribed to mental health challenges. One Washington implementer shared,

> "[Some clients] have mental issues, of varying forms. Whether it's, you know, a veteran that had PTSD, or whether it's a person that just has severe mental issues…We've worked on homes for people who are schizophrenic…You know they're at the lowest point in their lives sometimes…I've had several times where I've had to go out to these houses, and you know, talk to people…And I say, you know, 'everything's gonna be all right'… because everything seems like this mountain to them."

In this case, interviewees frequently indicated that issues not covered by Weatherization Readiness Funds, such as unsafe living conditions, hoarding, or various health and safety challenges, have increasingly become the primary reasons for deferrals.

Lastly, on the topic of serving and reaching eligible families, several implementers in Washington shared that they faced challenges reaching and serving Native American clients. In this case, implementers shared that there were either small percentages of Native American families in their service territory or they were making strides to strengthen connections with the Tribal organizations in their community.

3.4.2. Strengths/Desires

The greatest strength in both states is the presence of bilingual and culturally diverse staff. This improves communication with the families being served, enhances trust, and makes the weatherization process more efficient and accessible. Nearly every implementer interviewed in both states shared that they had bilingual staff and translated documentation. However, some implementers in Tennessee expressed frustration with

relying on phone translation services to talk to clients or having limited access to translators for intake documentation. One implementer in Tennessee shared,

> "Our dream situation is that we could have our application translated… to tell them [prospective clients] what's required and what kind of examples of documentation are actually eligible…So that way, when we advertise in a Latino newspaper, you can actually go and grab an application in your language and complete it, or have someone on staff that's bilingual. That would be awesome. Unfortunately, neither of us are bilingual, at least in Spanish. So yeah, it's hard to communicate instructions. Or you know who translate documents and stuff."

Lastly, and perhaps the most apparent sentiment among our interviews, was the desire to continue providing meaningful service to clients in need and improve the housing stock in their communities. Interviewees shared stories about clients whose lives were significantly improved by the WAP, resulting in energy savings and myriad non-energy benefits. For instance, an implementer in Washington shared,

> "There was a sweet lady…And when I went to her, she was wearing at least 3 or 4 pairs of pants and a coat, a hat, you know, snow boots in her house. Her husband passed away about a year before that, and he used to always keep the house warm with the wood stove-- She couldn't do it. She was in her eighties, and so she knew that she needed to keep her house about 50 degrees to keep her pipes from freezing. When I got there…It [the client's home] needed everything. There was no insulation anywhere near the house. So we air sealed. We did the attic…we did the underfloor. We did all kinds of stuff…And when we actually went back a few months after we were done, she was extremely comfortable and lived there the rest of her life. She's passed away, you know. But… There's something…from that job…kind of what I think of as 'energy takeback'. So it's now affordable for [client's name] to live comfortable in her house, and she's gonna take it back…and be comfortable, and not cold, because she can afford it…it's not like, 'if I turn it up to 72 degrees, where I'm comfortable, it's going to be $300, there's no way I can continue to live there', and she sees that coming-- It was like, 'Oh, I can afford the 80 to $100. And so I'm going to be comfortable in my house.'"

Similarly in Tennessee, one implementer described a conversation with a client that returned to thank them nearly a decade after receiving weatherization service,

> "I said, 'let me ask you, without this program, what would happen?' You know, they said, 'we don't know. We don't know whether we'd still be in our home, or what. Because this takes the burden off a lot of the things, [so] that we can do other things. You know that we can buy more food'… [redacted city] is, you know, it's a desert for fresh food. So they said, 'we can buy fresh food. We can eat. I can buy medicine.'"

4. Discussion

Our study used semi-structured interviews with WAP coordinators in nine CAAs serving the largest proportion of Latine households in rural Washington State and urban Tennessee. To supplement our interviews, we also analyzed federal and state-level policies for WAP implementation. Our results suggest significant differences in implementation at the local level, likely due to variations in state-level policies, geography, and local differences in population and housing stock.

4.1. Funding

Although Washington receives far more funds than Tennessee due to its additional state and utility funds, implementers from both states faced challenges associated with funding restrictions, such as administrative costs, timing of the funds, and blending multiple funding sources. As one interviewee described, these administrative requirements have been "layered over time", making program implementation increasingly difficult in recent years. Previous research has also documented these frustrations (18,41). Importantly, policy assessment of the ARRA period (2008-2014) suggests that the major influx of funds increased the political visibility of the program, which in turn increased both program oversight and programmatic costs (12). The layering of funding restrictions was felt especially among implementers serving communities with a significant proportion of impoverished families and poor housing stock rates. Often, coordinators dealing with high deferral rates or a limited number of certified contractors in their region reported dedicating significant time to coordinating contractor bids and processing income-eligible applications, which were unlikely to lead to a project.

When asked about funding strengths, interviewees in both states frequently shared that they felt their agency received adequate funding and had seen a marked increase due to the IIJA. Interestingly, there were important differences in utility partnerships in Washington and Tennessee. The primary difference in between the two states is the process of "leveraging" funds. Leveraging allows implementers to use multiple funding sources (e.g., DOE formula funds, Bonneville Power Administration [BPA] funds,

and utility funds) on one project (50). Because agencies in Washington receive utility funds as a supplement to their federal sources, they can leverage multiple funds in a single project, effectively increasing the flexibility and amount of funds per project. In FY2022, Tennessee was one of eight states that did not leverage funding sources, whereas Washington leveraged $19.7 million. Moreover, although Washington and Tennessee both contain federal agencies associated with power generation and distribution, the BPA and TVA function quite differently. For instance, CAAs in Washington receive direct agency funds through BPA, whereas TVA has started its own weatherization program, irrespective of the WAP called Home Uplift. Although the program provides additional weatherization services in Tennessee, it has increased competition for an already limited workforce in many regions.

The most discussed desire was funding flexibility so implementers could use their funds to hire staff, meet production goals, and realize community priorities for decarbonizing and maintaining healthy, affordable housing stock. Although the desire for funding flexibility is well-documented in the literature (11,18,20), our research supports the critical aspect of community-based approaches that CAAs use to identify and serve eligible families and the immense energy and non-energy benefits associated with the WAP.

4.2. Regulatory

The most commonly reported challenges by interviewees in both states were associated with evolving regulatory challenges. One specific regulatory challenge frequently referenced was the waiting list prioritization process, which often made clients feel that the process was not transparent. Although the DOE created priority requirements to ensure elderly and disabled clients were prioritized for service provision, implementers frequently shared that long waiting lists meant many families would be left unserved. One interesting idea that emerged from the interviews was integrating time into the priority points assignment, such as weighting the number of points per household by the number of months a family had been waiting for service.

Lastly, implementers shared that one strength of the program was their agencies' ability to identify and serve eligible clients. This aligns with previous research that suggests a strong demand for weatherization, especially when outreach is targeted and community-based (21). Using a community-engaged approach to enroll and assist eligible clients in completing paperwork could further boost application rates.

4.3 Market

The most critical market-related challenge discussed by interviewees was the lack of certified contractors in their area. This challenge was discussed most among WAP implementers in Tennessee. This finding is supported by recent survey research of WAP employees, which suggests that employee turnover among retrofit technicians and energy auditors is more than three times higher than the national average (52). Survey results indicated that the most common reasons for subcontractors' departures were lost production and time commitments (52). Moreover, when asked what the major barrier was to expanding training capacity among the weatherization workforce, 83% of respondents reported facing challenges finding qualified trainers (52). These results align with the desire that many implementers shared, which was for increased contractor certification opportunities. The call for more workforce training by coordinators in both states is especially interesting given Tennessee's ranking as the fastest-growing economy for jobs in the clean energy sector, whereas Washington is ranked 41st (53). In either case, the call for better workforce training could underscore Tennessee's potential in clean energy job growth and bolster Washington's commitments to clean energy jobs.

Another frequently discussed challenge was housing stock in the implementer's community. Although challenges associated with poor housing stock are well-documented in the literature (18,41), our study is likely the first to document the implementation of Weatherization Readiness Funds, which were generally viewed as a valuable resource and a powerful tool in reducing deferral rates. Lastly, an interesting aspect of this research was the frequency with which interviewees mentioned that weatherization helped to ensure housing for low-income families was safe and affordable, especially in a time when housing and

energy costs continue to rise. This has not been previously discussed in the literature, and there may be value in exploring both energy and housing justice in tandem in future studies.

4.4. Social

Nearly all interviewees mentioned challenges associated with reaching Latine families. Many coordinators shared challenges associated with a DOE policy implemented in 2016, limiting eligible families to ensure that one person in the household is a "qualified alien" (50). To our knowledge, previous literature on the WAP has not explored the impacts of immigration status on the program. Despite this, our research suggests that it may have significant ramifications for service provision, particularly in communities with high proportions of low-income Latine families. Furthermore, examining the service-seeking behaviors of Latine families will likely become increasingly important under the current administration, especially as anti-Latine rhetoric and threats of deportation rise. Notably, several coordinators highlighted that bilingual and culturally diverse staff and partner organizations were a key strength when building trust with Latine families. This protective relationship could be important for Latine individuals with limited English proficiency, who are more likely to experience higher risks of inadequate housing or infrastructure (33).

Another important finding was the importance of client ability in the provision of weatherization services. Often, client ability was the last barrier to servicing families and the most cited reason for deferral. Although client ability is understudied in the WAP literature, our research suggests that there may be an important opportunity to increase wraparound services to include mental health services.

4.5. Conclusion

Our research adds to the WAP literature by being the first to compare programmatic differences in WAP implementation across states and to explore the social challenges related to serving Latine families in different geographic contexts. Our study sets the stage for future studies, which could address some limitations to our findings, which arise from our small sample size and limited sampling frame. First, while our sample size is relatively small, it reflects rich data collected from our interviewees. Our study is the

first of its kind to engage with CAA members in more than one state qualitatively, making it a valuable multi-state case study of WAP implementation. Additionally, even though our findings are not representative of the more than 1,000 CAAs in the US, we have provided a replicable methodology that can be honed. Second, although our study aimed to understand the challenges and strengths of delivering the WAP among geographically diverse Latine populations, our sampling frame would have benefited from additional interviews with coordinators serving communities with smaller Latine populations to serve as comparison communities. Lastly, although in-person interviews allowed researchers to build connections with CAA implementers and see communities firsthand, recording in-person interviews was challenging, and unfortunately, one Washington interviewee's audio was damaged.

Our findings also suggest several exciting new ideas on the topic of equitable implementation of the WAP. For example, future lines of inquiry could interview low-income Latine families directly to explore the barriers to service-seeking, or partner with Latine-serving and Latine cultural organizations to understand how organizational coalitions might strengthen CAA relationships. Additionally, our research suggests that qualitatively understanding the meaningful impacts of weatherization on wellbeing, specifically mental health and those experiencing homelessness, could prove immensely helpful in understanding the breadth of non-energy benefits associated with weatherization. Lastly, questions of local workforce training programs in collaboration with community organizations could improve job opportunities in low-income communities and fill the need for additional workforce.

Our research highlights several significant challenges that have persisted throughout the WAP, including funding and regulatory restrictions, inadequate housing stock, and workforce issues. We identified several new strengths and challenges in the WAP, which likely reflect the critical inflection point in policy and funding changes that occurred before our interviews took place.

One of the biggest challenges that interviewees identified is the numerous regulatory hurdles that have been "layered over time," as one interviewee put it. In many ways, the WAP perfectly illustrates Klein and Thompson's (54) idea of "everything-bagel liberalism," which describes how progressives often design government programs with too many standards and rules, each of which appears sensible on its own, but

whose combined effects can unintentionally bring the program to a standstill. Over the past few decades, increased oversight and restrictions have made WAP coordinators' jobs more difficult, often leading to fewer homes being weatherized. In other words, if the WAP is to continue for decades into the future, it could benefit from a more streamlined focus on ensuring energy efficiency upgrades and fair implementation, which could be realized by supporting CAA coordinators who have the relationships and unmatched expertise needed to serve their communities.

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Figure_1

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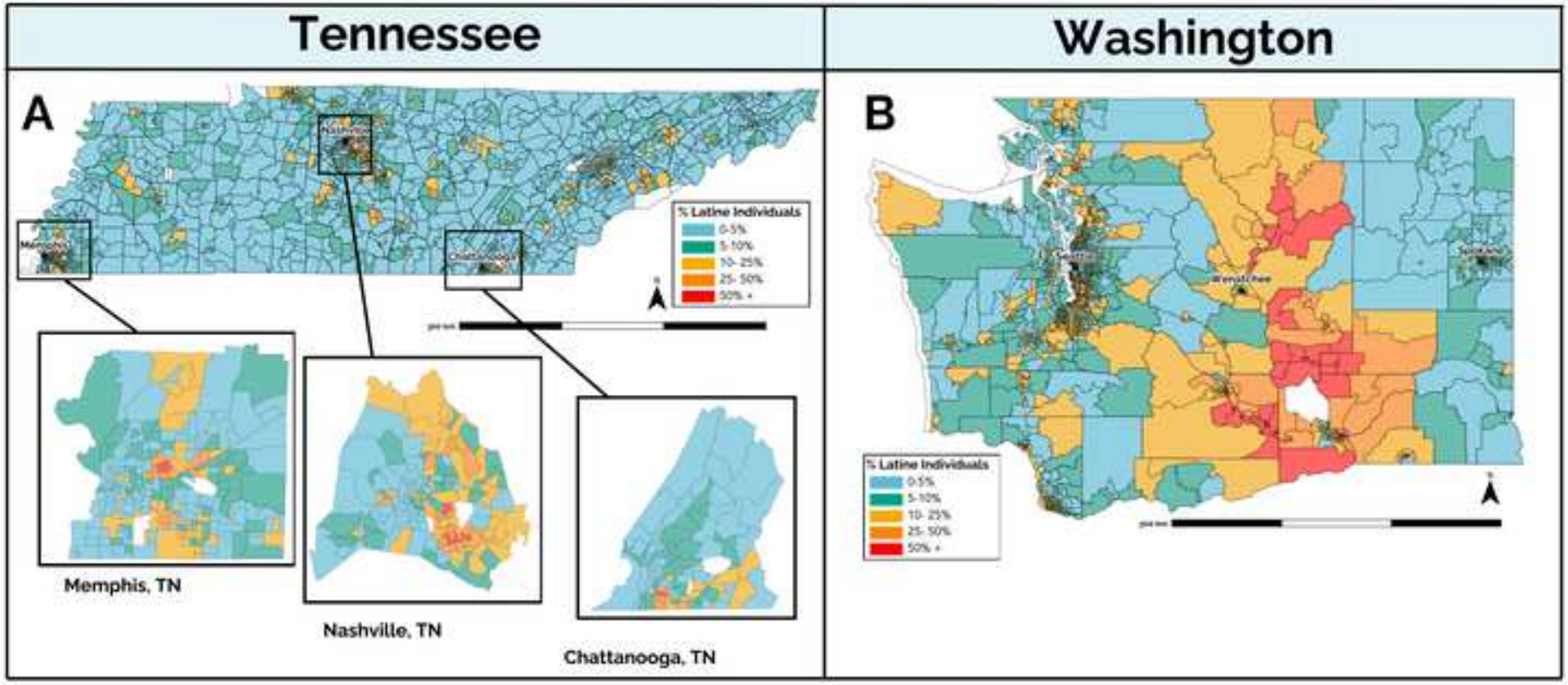

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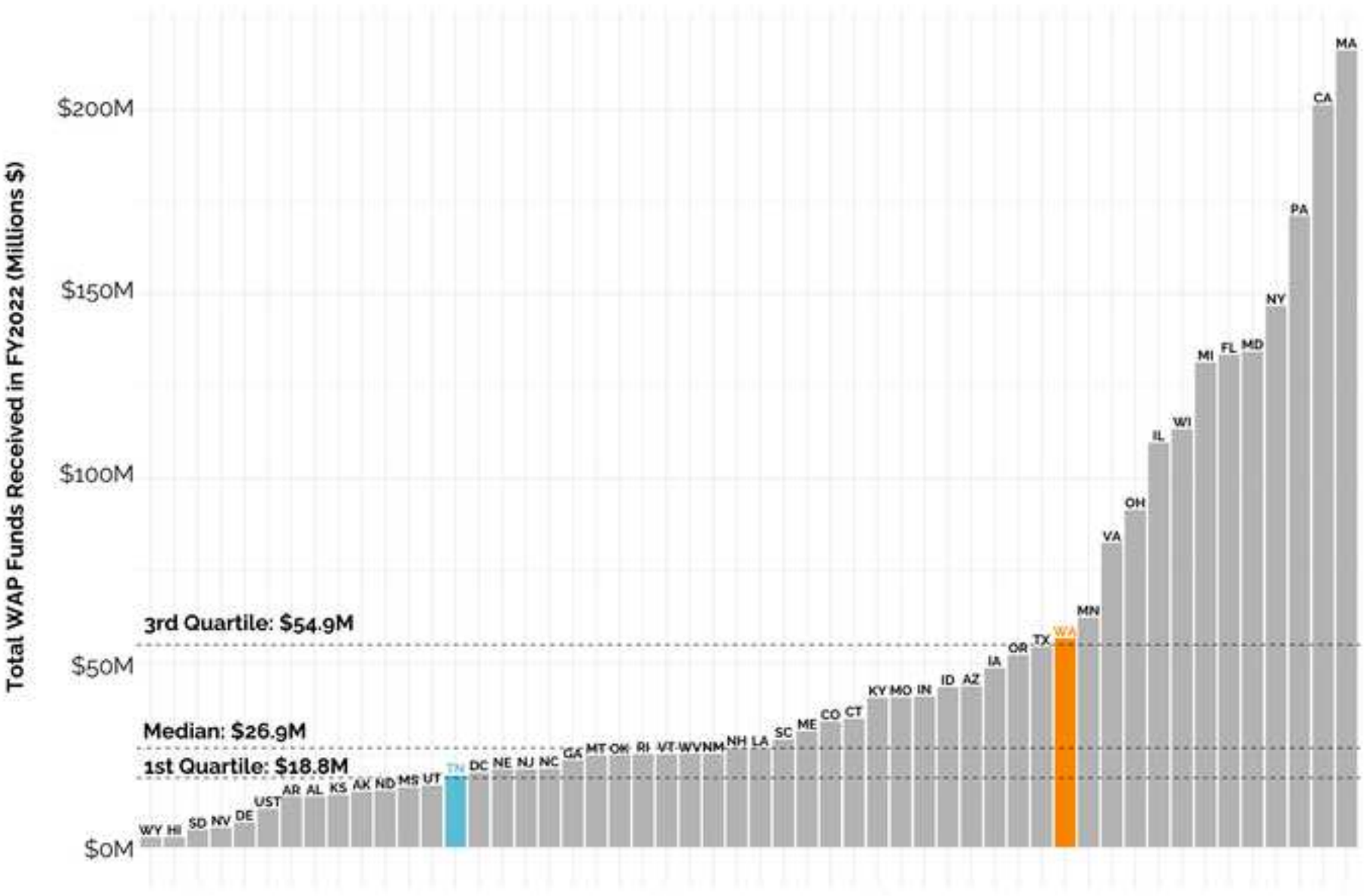

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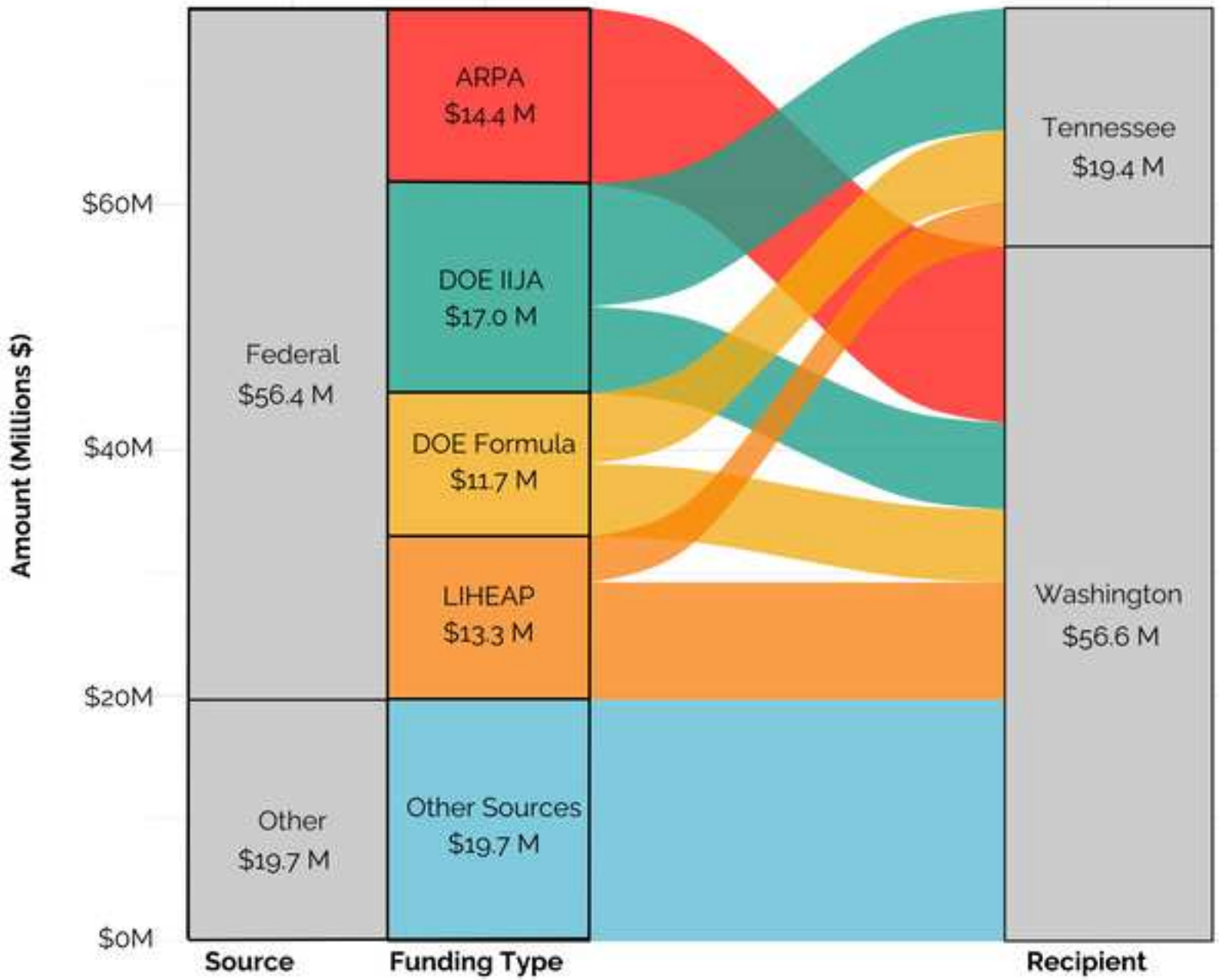

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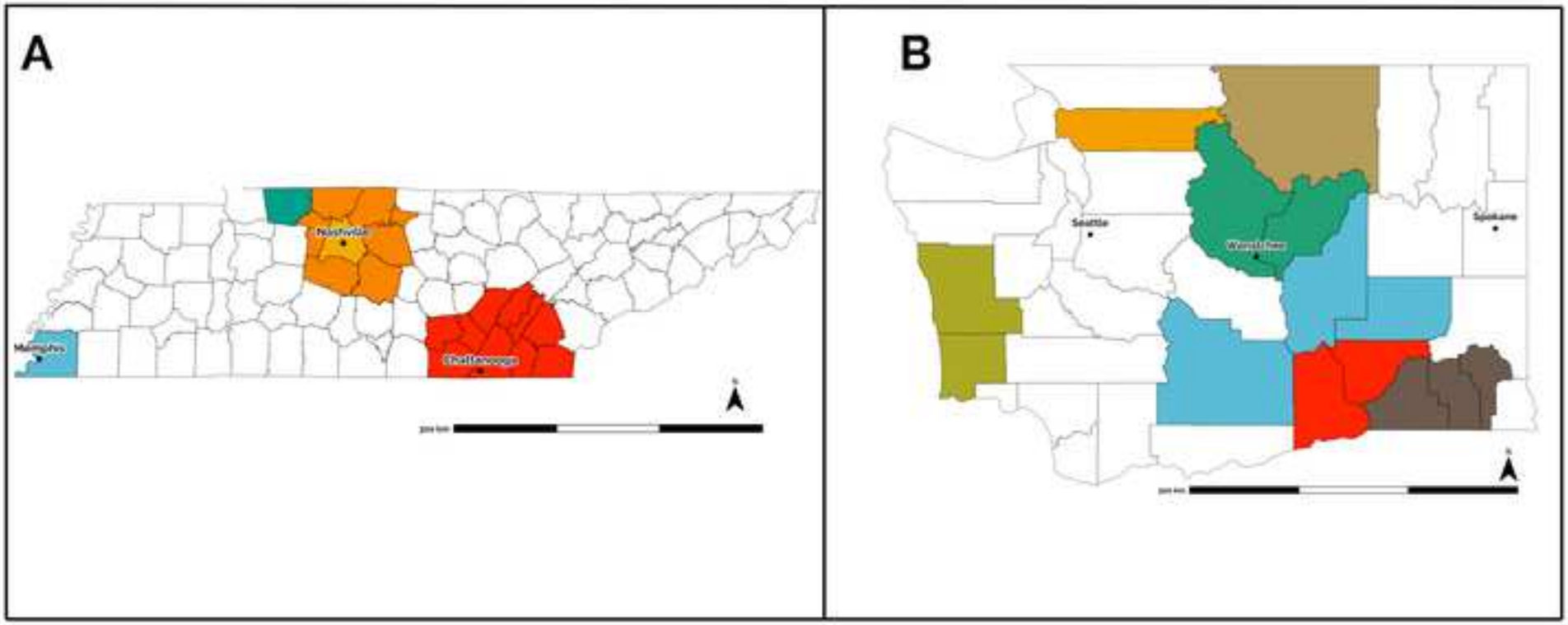

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FUNDING

Challenges

- Administrative Costs
- Timing of Funds
- Multiple Funding Sources
- Demonstrating Program Value

Strengths

- Adequate Funding Amounts
- Working with Utilities

Desires

- Funding Flexibility

REGULATORY

Challenges

- Evolving Regulatory Limitations
- Waiting List
- Invasiveness of Process
- Expansive Service Areas

Strengths

- CAA Partnerships
- Identifying Eligible Families

Desires

- Meeting Increased Quotas

MARKET

Challenges

- Limited Contractors
- Housing Stock
- Rising Costs

Strengths

- Landlord Support
- In-House Contractors (WA)
- Employee Retention

Desires

- Contractor Certification Opportunities

SOCIAL

Challenges

- Immigration Status
- Client Ability
- Serving Native American Families (WA)

Strengths

- Bilingual Staff
- Cultural Community Organizations

Desires

- Continued Meaningful Service

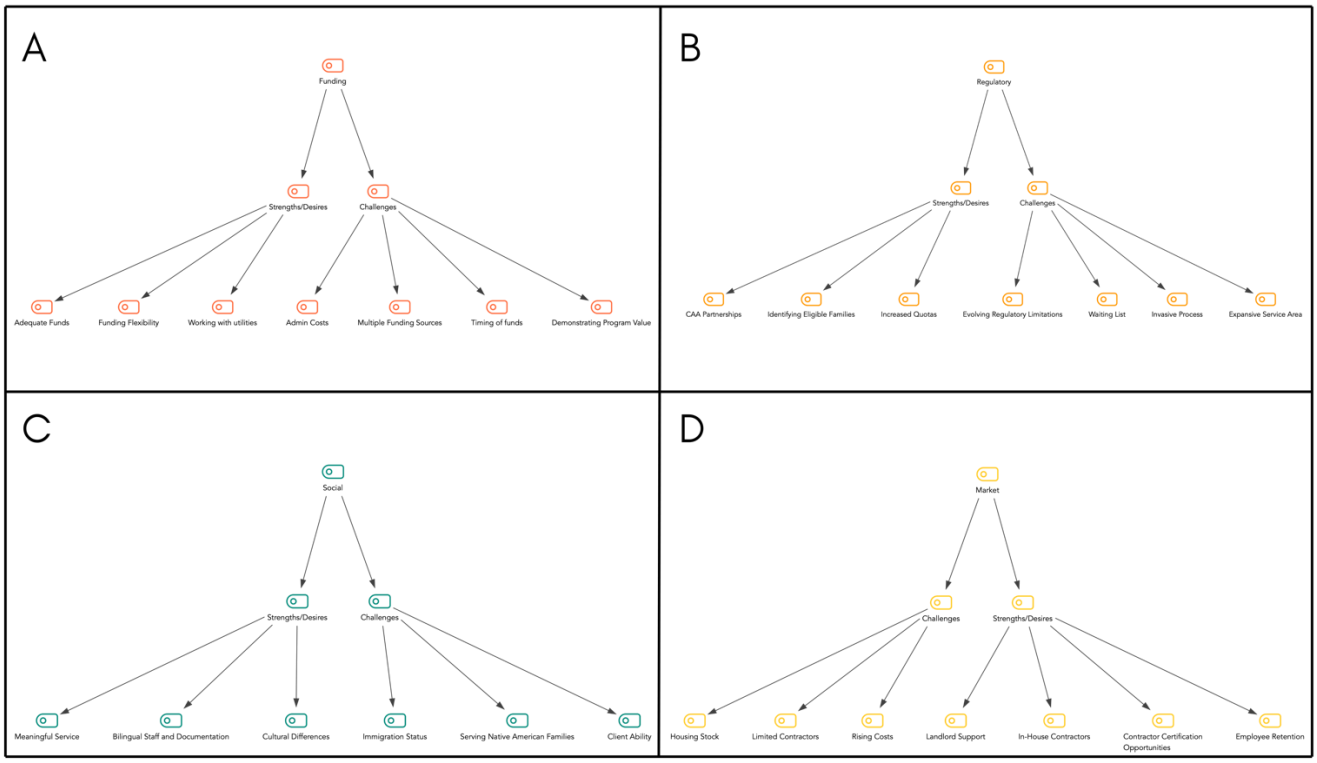
Supplementary Material Click here to access/download;Supplementary Material (Appendices);Supplementary_08_29.docx

Supplementary Table 1. Semi-structured interview guide used for interviews, including prompt, probe, and supporting literature used to develop the guide.

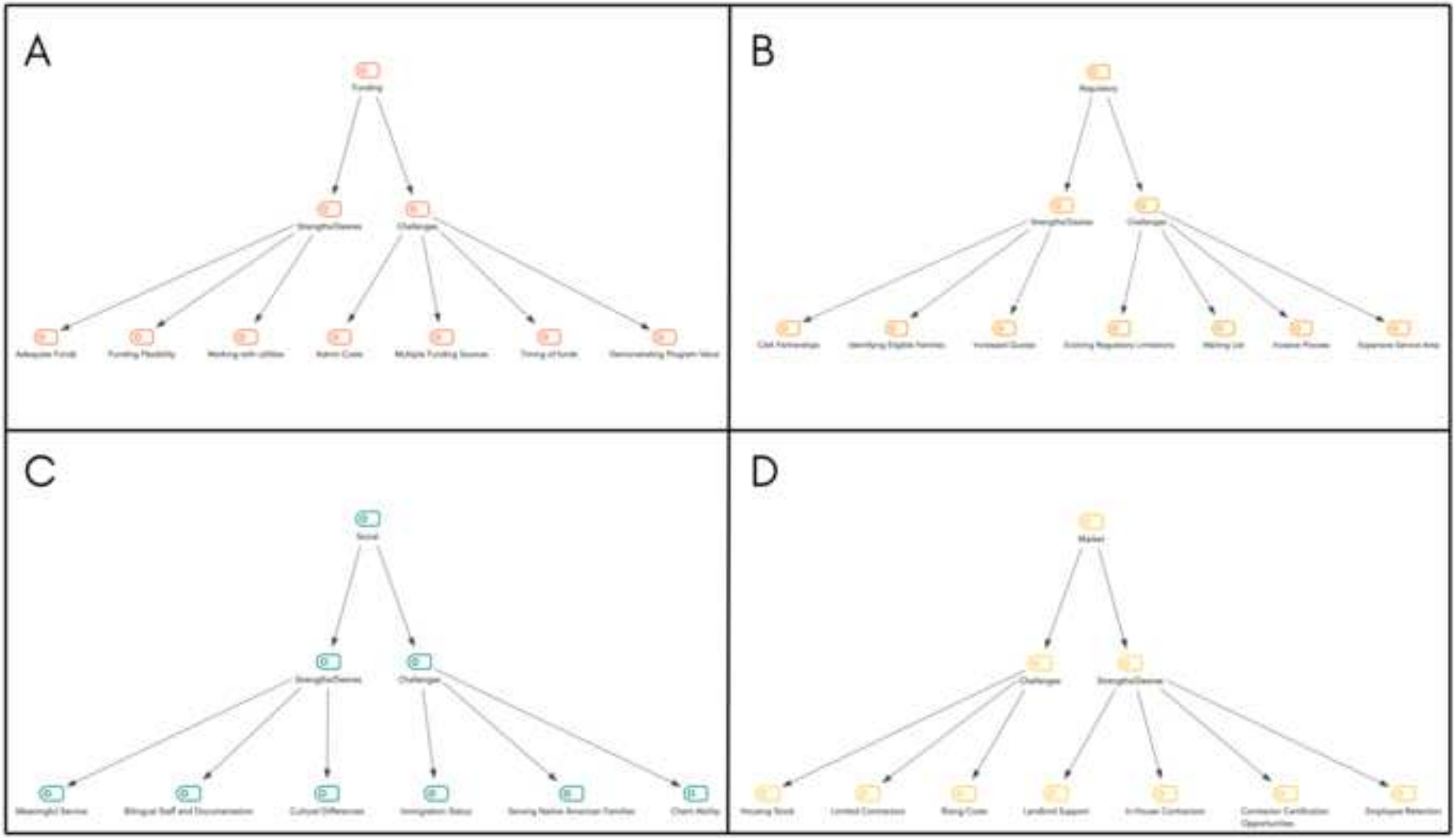
To start, I'd like to learn more about [CAA] and your role here.			
Theme	**Prompt**	**Probe**	**Supporting Literature**
Background	What is your role at [CAA]?	- How long have you worked there? - What does your job entail?	
	Can you tell me more about [CAA]?	- Roughly how many people work at [CAA]? - Do you work with other local organizations, such as local utilities or non-profits in your work?	
	What household energy services does [CAA] offer its clients?	- This could include things such as a weatherization assistance program, a utility shutoff assistance program, etc.	
Let's move on to discuss your thoughts and experiences about your [CAA]'s funding.			
Theme	**Prompt**	**Probe**	**Supporting Literature**
Funding/ Policy	Can you tell me more about how your agency is funded?	- Do you receive any federal and/or state funds for household energy services? - Do you receive any local funds or grants from other sources?	Raissi & Reames, 2020; Reames, 2016; Tonn et al., 2015
	Are there any challenges associated with receiving funding from these sources?	- Has your agency seen any significant changes in its funding structure due to federal allocations, such as the IRA and/or IIJA? - If you could change anything about the current funding structure for your agency's household energy services, what would it be?	Raissi & Reames, 2020; Reames, 2016
	Do you ever struggle to use all of your funds by the end of the fiscal year?	- Do you foresee not being able to fully utilize increased funds from the IRA and IIJA by 2031?	Caballero et al., 2022; Reames, 2016
Now I'll ask about your [CAA]'s eligibility process for receiving household energy assistance.			
Theme	**Prompt**	**Probe**	**Supporting Literature**
Regulatory	Can you tell me more about your process for identifying eligible homes?	- Do you prioritize any certain households (i.e., SNAP beneficiaries, elderly families, persons with disabilities, tribal families, or families with children, etc)? - How do you balance the costs of a retrofit and the proposed energy improvements?	Caballero et al., 2022, Tonn et al., 2015

	What is your CAA's process for household energy improvements?	- Approximately how many households can you improve in a year? - Approximately how much money do you spend on each home?	Tonn et al., 2015
	How often are households deferred in your service territory?	- What is the most common deferral reason? - If you could change one aspect of the eligibility process for home efficiency improvements, what would you change?	Tonn et al., 2015; Reames 2016; Caballero et al., 2022
Great, let's move on to discuss some of the market-related challenges related to [CAA]			
Theme	**Prompt**	**Probe**	**Supporting Literature**
Market	Do you ever face challenges serving renters in single-unit homes?	- In your experience, are landlords open to improving the efficiency of their units? - Do you find that homeowners are more or less willing to adopt energy efficient technologies than landlords? - Do you consider mobile homes single unit homes, and are there any unique challenges with serving families living in mobile homes?	Reames 2016; Caballero et al., 2022
	Great, and how about your experience serving families in multi-unit buildings, such as apartments?	- Are multi-unit landlords willing to improve the efficiency of their units? - Are there any local, state-level, or federal initiatives for multi-unit energy efficiency that you refer tenants and/or landlords to?	
	Are there any challenges specific to your region that make it challenging to provide your services?	- For example, is it difficult to maintain consistent workforce? - Is it challenging to help all the households in your service territory? - Are energy services especially costly, or perhaps less reliable? - Have you leveraged any local partnerships or opportunities to circumvent these challenges?	Caballero et al., 2022; Hardcastle, 2020; MacDonald et al., 2020
Thanks so much for this, the last topic we have is related to the social challenges of household energy efficiency.			
Theme	**Prompt**	**Probe**	**Supporting Literature**
Social	Is it challenging to identify and serve families in your service territory?	- For example, is it challenging to get families to sign up for WAP or LIHEAP services? - Are there approaches you've found to help reach and serve eligible families? - Research suggests that Latinx families are more likely to have received a shutoff notice or a struggle to pay their energy bills than white families, do you feel that this trend is similar for your agency?	Fowlie et al., 2018; Graff et al., 2021; Raissi & Reames, 2020

	Are there any challenges that you've found specifically when working with Latinx families?	- For example, is it challenging to translate materials, or have multi-lingual staff? - Are there any particularly effective approaches to reaching Latinx families your team has identified?	MacDonald 2020; Graff et al., 2021
Fantastic, we've reached the end of the interview. Is there anything else you'd like to add that we didn't cover? Or perhaps, any questions for us? *Thanks so much again for your time. We'd like to compensate you with a $50 gift card, and our department administration assistant will reach out to you via email to share it. We are also eager to share our final manuscript draft with you in the months coming, in hopes that you might have suggestions or find it useful. In the meantime, don't hesitate to email us if you think of any additional thoughts or concerns.*			



Supplementary Figure 1. Coding trees for thematic analysis, sorted by A) funding, B) regulatory, C) Social, and D) Market.

Supplementary Figure 1

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